systems in the realm of process control, radar systems, medical equipment or telecommunications.

Features of packet tracing/ analysis tools

There are a number of applications for which packet analysers or sniffers can be used in a constructive way. Given below is a list of the benefits of packet tracing tools:

- Analyse network problems
- Detect network intrusion attempts
- Detect network misuse by internal and external users
- Document regulatory compliance by logging all perimeter and endpoint traffic
- Gain information on network intrusion
- Isolate exploited systems
- Monitor WAN bandwidth utilisation
- Monitor network usage (including internal and external users and systems)
- Monitor data-in-motion
- Monitor WAN and endpoint security status
- Gather and report network statistics
- Filter suspect content from network traffic
- Serve as the primary data source for day-to-day network monitoring and management
- Spy on other network users and collect sensitive information such as login details or users' cookies (depending on any content encryption methods that may be in use)
- Reverse engineer proprietary protocols used over the network
- Debug client/server communications
- Debug network protocol implementations
- Verify adds, moves and changes
- Verify the internal control system's effectiveness (firewalls, access control, Web filter, spam filter, proxy, etc)

The open source packet analysis tools available are Wireshark, NetworkMiner and Snort.

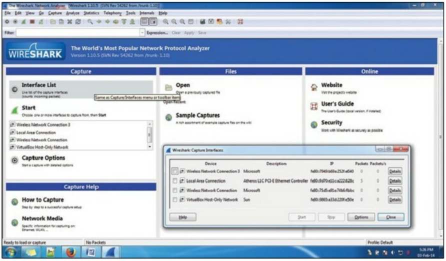

Figure 1: Selecting the interface list for packet analysis

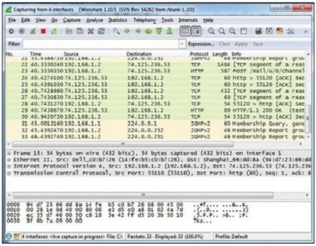

Figure 2: List of packets and related information analysed by Wireshark

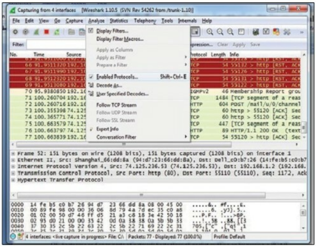

Figure 3: View enabled protocols for analysis

Wireshark

Wireshark is a free and open source network packet analysis tool. It is used for network troubleshooting, dissection, programming and communications protocol research, development and training. Initially, it was called Ethereal, and in May 2006, the venture was renamed Wireshark because of trademark issues. Wireshark is cross-platform. It runs on different UNIX-like frameworks including GNU/Linux, OS X, BSD and Solaris, and even on Microsoft Windows.

There is, likewise, a terminal-based (non-GUI) form called Tshark. Wireshark, and alternate projects distributed with it, like Tshark, are free software, released under the GNU General Public License.